

K-Means Clustering Techniques

Grouping Hospital Patients by Characteristics

Research Question

Is it possible to use known patient characteristics to group patients using K means clustering with centroids?

Analysis Goal

One goal of this data analysis is to build a useful grouping of clusters that hospitals can use to find an estimated range of initial days for some patients.

Method Justification

K means clustering begins by randomly assigning each data point to a specified K number of clusters, creating centroids within each cluster, representing the mean measurements of that cluster. The K means model then assigns each observation to the nearest centroid, based on Euclidean distance. Upon each observation grouping, both affected cluster's centroids are recalculated. This process repeats until all observations can no longer re-assign data points to closer groupings.

This analysis is expected to use continuous variables from the medical dataset to group patients into three to five centroids.

One assumption of K means clustering is that cluster sizes will be equally distributed among chosen K clusters.

Packages & Libraries

Sklearn.cluster KMeans , This will allow implementation of KMeans algorithm to determine the optimal cluster amount.

Pandas , Allows transformation of csv data into dataframe and manipulation of created dataframe.

Sklearn.preprocessing , Data scaling must be implemented to allow for clustering, this includes standard scaler which will do so.

Matplotlib.pyplot , Allows creation of scatterplot to visualize centroids and KMeans elbow method on a line chart.

Seaborn , Allows creation of scatterplot using dataset variables.

Sklearn.metrics Silhouette_score , This allows for a model silhouette score to evaluate the clusters.

Data Preprocessing Goal

The dataset will need to be standardized using standard scaler before running the K means algorithm.

Variables

The chosen dataset variables will be the continuous variables, 'Income', 'VitD_levels', and 'Initial days'.

Data Preparation Steps

Lines 9-11: First, the data will be loaded from the csv file into a pandas dataframe.

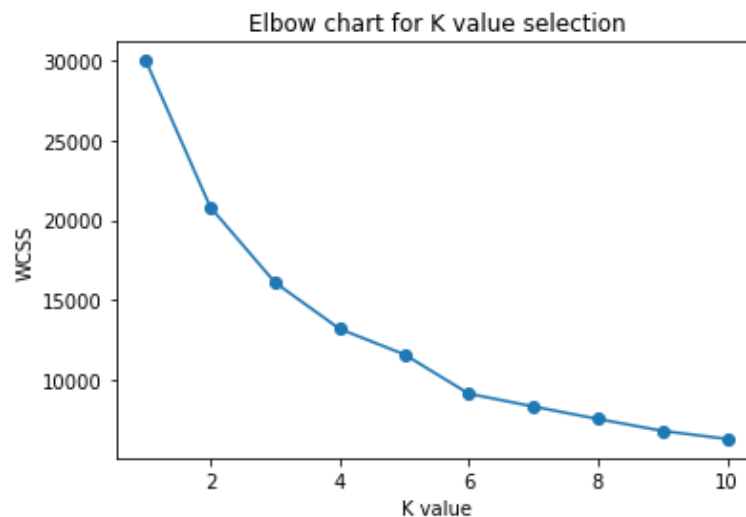
Lines 13-14: A new dataframe labeled 'X' will be defined to store only the columns which will be used in cluster analysis. This step will also round numeric variables.

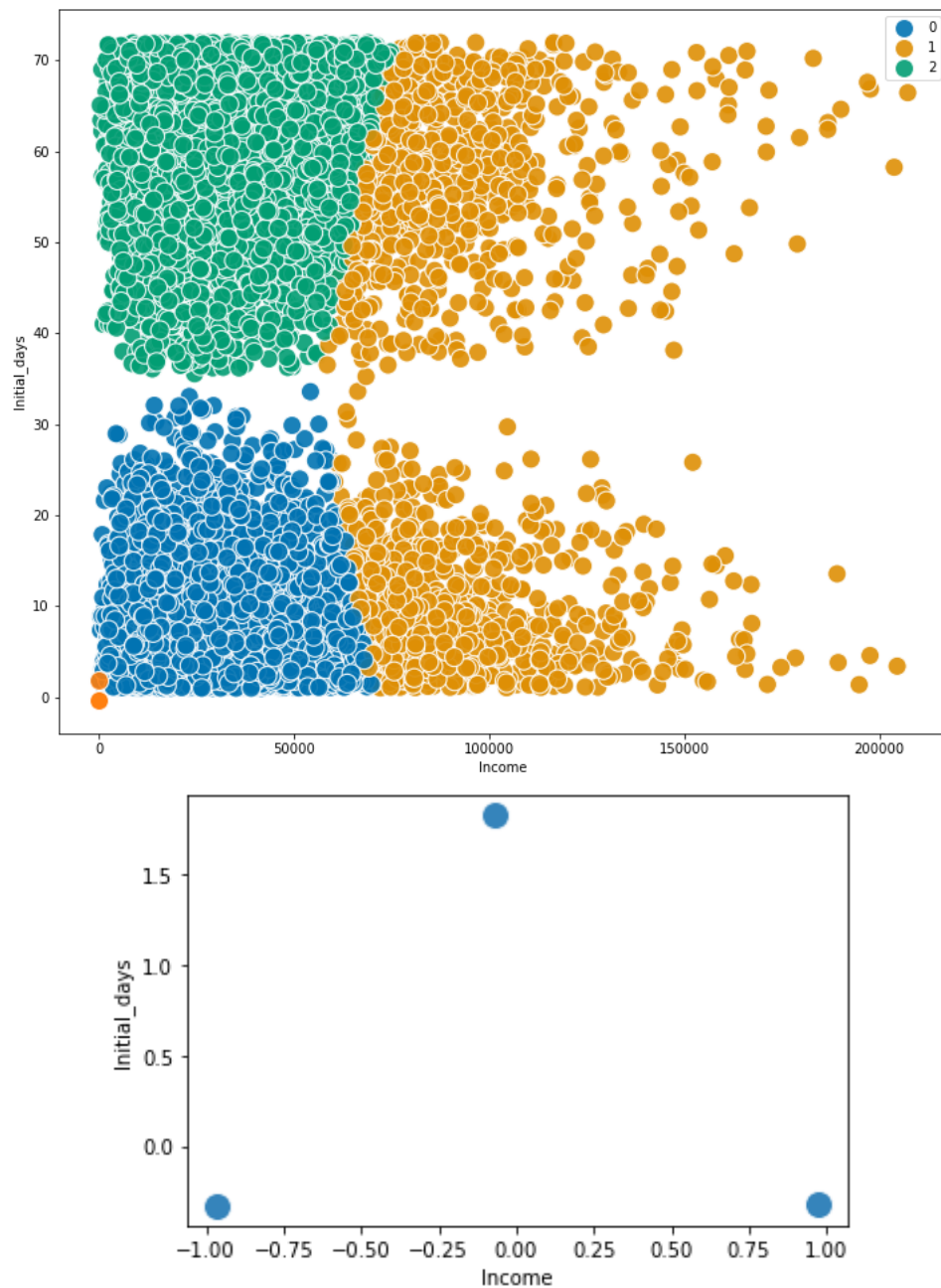
Lines 16-18: The newly defined X dataframe is now scaled using standard scaler and the scaled dataframe is given a unique name.

Lines 24-29: KMeans algorithm is implemented to assess the optimal K number of clusters.

Optimal Number of Clusters

The optimal K number of clusters was recommended by elbow method to be set at three. The implemented elbow method works by creating a visualization of the within cluster sum of squares at each possible K value in a selected range, allowing users to visually see the 'elbow', the point where the rate of change in WCSS takes a sharp decrease. This point indicates diminishing returns in increasing K values and can therefore be selected for cluster analysis.





Cluster Quality

This analysis suggests that patients be classified into the following clusters:

Cluster 0: Patients with low income and low initial days.

Cluster 1: Patients with all ranges of income and average initial days.

Cluster 2: Patients with high income and low initial days.

Results & Implications

This clustering analysis has suggested that hospitals group patients into one of three clusters, low income & low initial days, the whole range of income & average initial days, and high income low initial days.

The silhouette score of 0.34 indicates that the clustering model is far from the perfect score of 1 which would indicate that clusters have no overlap and can always successfully predict which cluster patients would be grouped into, but there is enough validity to the model that it can be used to provide a rough prediction for initial days based on cluster.

Limitations

One limitation of this data analysis is that clustering is only implemented on numeric variables, not considering any of the information provided by categorical data.

Course of Action

Based on the findings from this Kmeans clustering around centroids analysis, it would be recommended that hospitals apply this clustering model on future patient data during their stay to estimate the range of necessary initial days.